

Embedded Linux & System Software Interview Handbook

Chapter 4 - Board Bring-up (Part 2)

Topics: Power Rails, PMIC, Reset Signals, Clock Initialization and DDR Bring-up.

4.6 Power Rails

The first task after receiving a board is verifying that every required voltage rail is present. Typical rails include Core (0.8–1.0V), DDR (1.1/1.2V), IO (1.8V), and Peripheral (3.3V). Use a multimeter for static checks and an oscilloscope for sequencing and stability.

Interview Tip: Never start debugging software until power rails are verified.

Power-Up Sequence

Adapter/Battery → PMIC → Core Voltage → DDR Voltage → IO Voltage → Peripheral Voltage → CPU Reset Release

4.7 PMIC (Power Management IC)

The PMIC generates and sequences multiple voltage rails. It may also monitor thermal events, battery status, watchdog timers and power buttons. If a required rail is missing, the CPU will never execute BootROM.

Debug Checklist

- Verify PMIC enable pin.
- Check PGOOD (Power Good) signals.
- Measure output rails.
- Confirm enable sequence from schematic.

4.8 Reset Signals

After power becomes stable, reset signals must be released. A stuck reset line keeps the processor permanently halted.

Important reset sources include CPU reset, watchdog reset, external reset button and brown-out reset.

4.9 Clock Initialization

No processor can execute instructions without a valid clock. Verify the crystal oscillator, PLL lock status and derived clocks for CPU, DDR and peripheral buses.

Typical Clocks

24 MHz crystal → PLL → CPU Clock

PLL → DDR Controller

PLL → UART/I2C/SPI

Interview Question

Why does UART print garbage characters?

Possible causes include incorrect baud rate, unstable clock source or wrong UART clock configuration.

4.10 DDR Bring-up

DDR initialization is one of the most critical bring-up stages. SPL or firmware configures the DDR controller, performs timing configuration and memory training before U-Boot loads into external RAM.

Common DDR Failures

- Wrong timing values
- Incorrect voltage
- PCB routing issues
- Memory training failure
- Wrong DDR type or density

Samsung Example

During TV platform development, engineers often confirmed DDR initialization by checking whether U-Boot successfully relocated itself into external RAM. If relocation failed, DDR initialization became the primary debugging focus.

Practical Debugging Flow

1. Measure power rails.
2. Verify PMIC outputs.
3. Check reset deassertion.
4. Verify oscillator and PLL.
5. Capture UART logs.
6. Confirm DDR initialization.
7. Verify BootROM handoff to SPL/U-Boot.

Interview Questions

1. What does a PMIC do?
2. Why are power sequencing and reset timing important?
3. How would you debug a board with no UART output?
4. Why is DDR training required?
5. What tools do you use to validate clocks?

Key Takeaways

Power, reset, clock and DDR form the foundation of board bring-up. If any one of these fails, Linux will never boot. Experienced engineers always validate hardware fundamentals before debugging software.